

EDITORIAL

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Research attitudes at a crossroads: advancing research on smart and sustainable cities

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The educational paradigm embraced in society for decades has predisposed learners to mechanical thinking patterns. There are obvious differences between the research attitudes of the pioneers of the “Scientific Revolution” and that of today’s researchers, who are no strangers to the modern education system. Single-discipline-centred research tends to restrain the learners’ creativity and beliefs about cognitive abilities. How can the adoption of more ambitious research strategies (e.g., multidisciplinary schemes) impact scientific innovation? What role will *Smart Construction and Sustainable Cities* (SCSC) play in promoting multidisciplinary research?

1 The original intention of the research matter

Like their medieval predecessors before them, the pioneers of the “Scientific Revolution” [1] such as Copernicus, Galileo, and Isaac Newton, fell into science by interest and the want to understand the world around them. Several of these scientists initially attended university with the intent of absorbing and learning the liberal arts (including astronomy, geometry, and arithmetic) from their predecessors. For instance, Newton attended the Trinity College of Cambridge in the early 1660s [2]. These scholars, however, regarded the knowledge taught by these institutions as old-fashioned. They, therefore, began developing new ideas and methods that would become unrivalled contributions to science. For

example, Newton developed calculus to tackle the issues he encountered in his physics research. He later developed the law of gravitation, which is one of the most remarkable discoveries from the 1660s to the present time.

The common thread among the pioneers of the Scientific Revolution is the practice of education and research, with interest as the primary motivation. This interest promoted the concomitant exploration of various disciplines by scholars around that period. Karl Terzaghi, the father of modern soil mechanics, founded the discipline based on the aforementioned interest-motivated research. He originally studied mechanical engineering at a university which provided him with a sound foundation in mechanics [3]. Terzaghi was also fascinated by geological science and, at some point, combined these two disciplines (explaining geological phenomena from a mechanical perspective) to create soil mechanics [3]. He subsequently developed additional futuristic concepts for geotechnical engineering. These examples demonstrate that the original intention of research should not be single-focused *i.e.*, constrained by professional or academic disciplines. What made early scientists successful was their practice of curiosity-driven research and the exploration of various disciplines. However, with university development, discipline-focused curricula have gradually been adopted by higher education systems worldwide.

2 Mono-discipline oriented research: a myth or an opportunity

From the early modern period to the twentieth century, Western universities have substantially evolved in terms of their structure and functioning. For instance, the

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twentieth century brought about a dichotomy between science and technology, developing into different disciplines, such as geotechnical engineering, structure engineering, engineering management, water conservancy engineering, and traffic engineering. This differentiation among disciplines promoted the systemic training of graduates based on discipline rather than interest (in contrast to the research attitudes discussed previously). This shift in the educational paradigm has generated changes in the attitudes of learners/scholars towards research. Due to the uniqueness of the research focus, for example, graduate students are very likely to have a discipline-constrained perception of their research, which, combined with potential suboptimal tutoring, seldom produces innovative achievements. In reality, early research attitudes have been lost over the decades due to the evolution of educational structures and systems. It is speculated that the creative abilities of early researchers comprised short paths between concepts and high connectivity. Individuals with higher creativity generally demonstrate a flexible network structure indicative of creative thinking [4]. This characterisation of creative thinking and semantic memory contrasts, for example, with the current doctoral training system, which tends to convey little research, thereby limiting the creative thinking of the learner. This state of play substantiates the relative stagnation of certain disciplines (e.g., soil mechanics and physics) that have witnessed no significant breakthroughs in decades or even centuries. The lack of discoveries has resulted in an endless stream of practical engineering problems, which existing theories have failed to accurately predict or solve.

3 A needed resurgence of early research attitudes

While the development of modern technology has opened up new opportunities for solving long-deemed indecipherable problems, the resurgence of ancient research attitudes is bound to hinder the dynamics of innovation and discovery. It is critical to recourse to research approaches that promote an interdisciplinary exploration of targeted mechanisms, to regain the “origin of research”. The origin or original intention of research should be free from the limitations of discipline and direction. An effective method for scientific research is a combination of inductive and deductive reasoning. The reader can refer to Nola [5] for a detailed introduction to these methods.

As a matter of illustration, in the early 1920s, Terzaghi used his personal perception or “art sensing” to propose the assumption of effective stress based on his long-term “live with the soil” (means one must love and understand every aspect of soil behaviour both in the laboratory and in the field) [3]. Since Terzaghi hypothesised the

concept of effective stress in 1923, macroscopic measurement of this quantity has always been achieved using an indirect method. Specifically, the effective stress has been obtained by subtracting the measured pore water pressure from the measured total stress. Although this approach has been deemed trustworthy, the development of a direct measurement method has remained at an embryonic stage for decades owing to innovation difficulties and stagnation. With the development of optical fibre technology, this centennially unsolved case has been effectively resolved in recent years by Hong Kong-based researchers [6]. Their work provided experimental confirmation of Terzaghi’s hypotheses. This contribution was an unprecedented achievement in the field of soil mechanics following Terzaghi’s seminal work. The development of modern information perception technologies, e.g., sensors and digital technologies is bound to upgrade our early “art sensing” and promote the development of innovative construction techniques that preserve the sustainability of modern cities. Some examples include good results obtained via the observation of field disc cutter wear [7] and shield vibration [8] to judge the geological characteristics and timing of the cutter. More construction technologies have been developed to guide intelligent compaction processes and quality control using global positioning systems [9]. Data-driven methods are also applied in underground space [10] and disaster prevention in urban areas [11]. Moreover, green material [12] for sustainable cities is another aspect of the needs.

4 Role of smart construction and sustainable cities

The journal *Smart Construction and Sustainable Cities* is committed to the promotion and interdisciplinary research that embraces the philosophy discussed above. This provides an interdisciplinary forum for applying information technologies to the construction and management of sustainable cities.

Cities are economic powerhouses that serve as nerve centres for the global economy and society and have the potential to be agents of positive change. However, rapid and disorderly urbanisation has led to deep inequalities and an unsustainable urban environmental reality. The challenges posed by the urbanisation trend are numerous, spanning social, economic, and environmental issues that are intertwined. Insecurity, public health, inequality, pollution, waste management, and accessibility of services are responsible for the well-being and development of people. Making cities more inclusive, safe, resilient, and sustainable, as called for in SDG11, is a timely and pressing necessity. The development of smart cities is a rational strategy to achieve this. The Fourth Industrial Revolution has generated powerful technologies that are pivotal game-changers. *Smart Construction*

and Sustainable Cities offer a forum for the discussion of intelligent solutions to achieve smart governance of societies and self-adaptive and sustainable cities. More importantly, the journal embraces a multidisciplinary vision towards valuable research/scientific achievements in the application of information science and technology for the development of smart and sustainable cities. Therefore, in line with the previously discussed ‘research attitudes,’ *Smart Construction and Sustainable Cities* welcome research from various disciplines, including civil engineering, artificial intelligence, urban planning and management, manufacturing, urban governance, data science, transportation, mechanical engineering, electronic information, and computer science.

The journal will feature transformative solutions, systematic approaches, science-policy-practice, and innovative systems to support the development of urban intelligent construction and maintenance, intelligent manufacturing, intelligent perception and digital image processing, the future urban brain, and intelligent computing. Furthermore, in support of innovation and novelty, the journal has waived article processing charges. The authors, therefore, do not need to pay any article processing charges for its early issues. The journal strongly welcomes stimulating comments, reviews, perspectives, case studies, communications, and original research falling within the scope of the journal and in support of the Sustainable Development Goals, particularly SDG11.

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References

- Wootton D (2017) History: Science and the Reformation. *Nature* 550:454–455. <https://doi.org/10.1038/550454a>
- Moore JC (2019) A Brief History of Universities. Palgrave Pivot Cham, London. <https://doi.org/10.1007/978-3-030-01319-6>
- Goodman RE (1999) Karl Terzaghi: the Engineer as Artist. *American Society of Civil Engineers (ASCE)*, Reston, p340
- Denervaud S, Christensen AP, Kenett YN et al. (2021) Education shapes the structure of semantic memory and impacts creative thinking. *npj Sci Learn.* 6:35. <https://doi.org/10.1038/s41539-021-00113-8>
- Nola R (2007) Theories of Scientific Method: An Introduction. Acumen Publishing, Stocksfield.
- Yin JH, Qin JQ, Feng WQ (2020) Novel FBG-based effective stress cell for direct measurement of effective stress in saturated soil. *Int J Geomech* 20(8):04020107
- Zhang N, Shen SL, Zhou A (2023) A new index for cutter life evaluation and ensemble model for prediction of cutter wear. *Tunn Undergr Space Technol.* 131:104830. <https://doi.org/10.1016/j.tust.2022.104830>
- Shen SL, Yan T, Zhou A (2023). Estimating locations of soil-rock interfaces based on vibration data during shield tunnelling. *Autom Constr.* 150:104813. <https://doi.org/10.1016/j.autcon.2023.104813>
- Yao YP, Song EB (2023). Intelligent compaction methods and quality control. *Smart Constr Sustain Cities.* 1:2. <https://doi.org/10.1007/s44268-023-00004-4>
- Peng FL, Dong YH, Wang WX, Ma CX (2023) The next frontier: data-driven urban underground space planning orienting multiple development concepts. *Smart Constr Sustain Cities.* 1:3. <https://doi.org/10.1007/s44268-023-00003-5>
- Chai JC, Wu HZ (2023) Prevention/mitigation of natural disasters in urban areas. *Smart Constr Sustain Cities.* 1:4. <https://doi.org/10.1007/s44268-023-00002-6>
- Dagliya M, Satyam N, Garg A (2023) Desert sand stabilization using biopolymers: review. *Smart Constr Sustain Cities.* 1:5. <https://doi.org/10.1007/s44268-023-00001-7>